

The Dodecahedral Dice Challenge Solution

We have 4 dodecahedral dice (each with 12 faces), labeled Blue, Red, Green, and Black.

Each die has unique numbers from 1 to 48 across all dice. The dice must be balanced so that:

- Any 2 dice have equal probability of rolling the higher number.
- Any 3 dice have equal probability of rolling the highest number.
- Any 4 dice have equal probability of rolling the highest number.

A valid construction can be achieved by an extension of Efrons nontransitive dice design, generalized for 4 dice with 12 faces each.

Blue Die:

1, 8, 9, 16, 17, 24, 25, 32, 33, 40, 41, 48

Red Die:

2, 7, 10, 15, 18, 23, 26, 31, 34, 39, 42, 47

Green Die:

3, 6, 11, 14, 19, 22, 27, 30, 35, 38, 43, 46

Black Die:

4, 5, 12, 13, 20, 21, 28, 29, 36, 37, 44, 45

Each die contains one number from every group of four consecutive integers

1 thru 4 5 thru 8 etc.

This ensures that over all comparisons, each die has an equal probability of producing the highest roll.

For example:

- In any pairwise comparison, each die wins exactly half the time.
- In any 3-way or 4-way comparison, all have equal expected rank.

Thus, these dice satisfy the fairness constraints.